SPLUNK AGENTIC OPS HACKATHON · OBSERVABILITY

L LLMWatch

Your LLM got worse last night.
Splunk saw it — and rolled it back.

Agentic Ops | HEC · SPL · Dashboards | MCP Server | Hosted Models

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THE PROBLEM

LLM quality degrades silently

A model update, a prompt change, or a drifting RAG index drops answer quality 20–50% — with **no error, no red dashboard, no alert.**

0
alerts fire

Standard monitoring watches latency and errors — never *groundedness*.

days
to detect

Teams find out from customer complaints, not telemetry.

–52%
real regression

A single model bump silently halved answer quality in our live test.

The metric that matters is never monitored. Splunk already watches infrastructure at scale — point it at **answer quality**.

THE SOLUTION

An agent that runs on Splunk

LLMWatch instruments every LLM call into Splunk, then closes the loop — it doesn't just alert a human, it **remediates.**

SENSE read Splunk (MCP/REST) → **DETECT** vs 24h baseline → **INVESTIGATE** hosted-model root cause → **DECIDE** rollback / reroute → **ACT** + approval gate → **LOG** audit to Splunk

Passive dashboards stop here

“Turn a card red, email ops-team@company.com.” The human still has to notice, investigate, and fix.

LLMWatch keeps going

Root-causes with a Splunk hosted model and rolls back the bad model — behind a human-approval gate for safety.

HOW WE USE SPLUNK

Five Splunk capabilities, one loop

CAPABILITY	HOW LLMWATCH USES IT	PRIZE
HEC	Privacy-safe ingestion of every LLM call (prompts hashed)	Core
SPL	Quality trends, regression detection, hallucination rate, cost/quality	Core
MCP Server	Agent reads Splunk securely over MCP (Cloud) — REST on local Enterprise	Best Use of MCP
Hosted Models	`gpt-oss` as LLM-as-judge *and* root-cause analyst	Best Use of Models
Dashboard Studio	Real-time Quality Observatory + Regression Investigation	Design

One project, eligible for the **Observability** prize + two bonus prizes.

PROOF — VERIFIED LIVE

Not a mock. Real Splunk.

Seeded events via HEC, ran the agent against live SPL on a Splunk Enterprise instance:

```
SENSE · pulled 2 model signals from Splunk [REST]
DETECT · gemini-2.0-flash-v2.3 dropped 52.1% (0.84 → 0.402) — REGRESSION
INVESTIGATE · gpt-oss root cause: authentication/authorization [HIGH] → rollback
ACT · executed: Active model v2.3 → v2.2. Traffic restored.
LOG · decision written to index=llmwatch (audit trail)
```

```
// read back out of Splunk — sourcetype=llm_agent_actions
{ "status":"regression_remediated", "action.kind":"rollback",
  "result.detail":"Active model v2.3 → v2.2. Traffic restored." }
```

THE QUALITY METRIC

Groundedness, scored 0–1

LLM-as-judge via a Splunk hosted model. Validated against a labelled benchmark:

WITH RETRIEVAL CONTEXT

0.621

Grounded, cites concrete specifics.

WITHOUT CONTEXT

0.158

Generic, hedging, hallucination-prone.

This one score feeds **both** Splunk anomaly detection **and** the agent's detection step. 3.9× quality lift from context — measured, not claimed.

ARCHITECTURE

HEC in · agent loop · audit out

LLM apps — any provider
instrumented · prompt hashed · privacy-safe

▼

Collector → **HEC** → `index=llmwatch` → Splunk Enterprise / Cloud

▼

LLMWatch Agent

SENSE → DETECT	read Splunk via MCP Server / REST · SPL vs 24h baseline
INVESTIGATE	Splunk hosted model roots-causes the failing calls
DECIDE → ACT	rollback / reroute · human-approval gate
LOG	audit decision → `index=llmwatch`

▼

Dashboards · Alerts · PagerDuty / Slack

WHY IT WINS

Judged on four criteria

Technological Implementation

Real HEC + MCP + hosted model + a working control loop. Verified live, 9 passing tests. Run it in 5 seconds.

Quality of the Idea

Meta-observability for LLMs — the groundedness metric nobody monitors. Novel and on-trend.

Potential Impact

Every team shipping LLMs has this blind spot. Catches silent drift before customers do.

Design

Enterprise dark Quality Observatory in Dashboard Studio. Demos beautifully.

Agentic, not passive — it *acts*, with responsible-autonomy safeguards.

RESPONSIBLE AUTONOMY

It acts — but it shows its work

🔒 Approval gate

Destructive actions (rollback, reroute) wait for one human click unless explicitly autonomous.

🧾 Full audit trail

Every decision logged back to Splunk — `llm_agent_actions` — queryable and reviewable.

🛡️ Privacy by default

Prompts hashed before ingestion. Secrets env-only and masked in logs.

The agent earns trust the way an SRE does: **transparent decisions, reversible actions, a gate on anything destructive.**

PRODUCTION AI, OBSERVED & DEFENDED

LLMWatch

Splunk scale. LLM-as-judge groundedness. One agent closing the loop.

HEC | SPL | MCP Server | Hosted Models | Dashboard Studio | Alerts

`github.com/manojmallick/llmwatch`

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